

3 (long) The input characteristic of an n-channel enhancement-mode field effect transistor is shown in Fig. 3a. The supply voltage, V_{DD} is 10 V and $R_1 = 4.25 \text{ M}\Omega$.

(a) The transistor is to be used at an operating point set as $V_{DS} = 5 \text{ V}$, $V_{GS} = 1.5 \text{ V}$ and $I_D = 4 \text{ mA}$, in a common-source configuration as shown in Fig. 3b. Determine the values of the resistors R_2 and R_3 required to achieve the desired operating point, stating any assumptions made. [4]

(b) At this operating point, the small-signal drain resistance $r_d = 10 \text{ k}\Omega$. From the graph in Fig. 3a, determine the small-signal mutual conductance, g_m . [6]

(c) Draw the small-signal equivalent circuit of the amplifier with the salient features clearly labelled. [3]

(d) Calculate the input resistance, output resistance and voltage gain of this circuit. [12]

(e) A voltage of amplitude 400 mV from a signal source with internal resistance $100 \text{ }\Omega$ is connected to the input terminals of the amplifier. Calculate the resulting output voltage. [5]

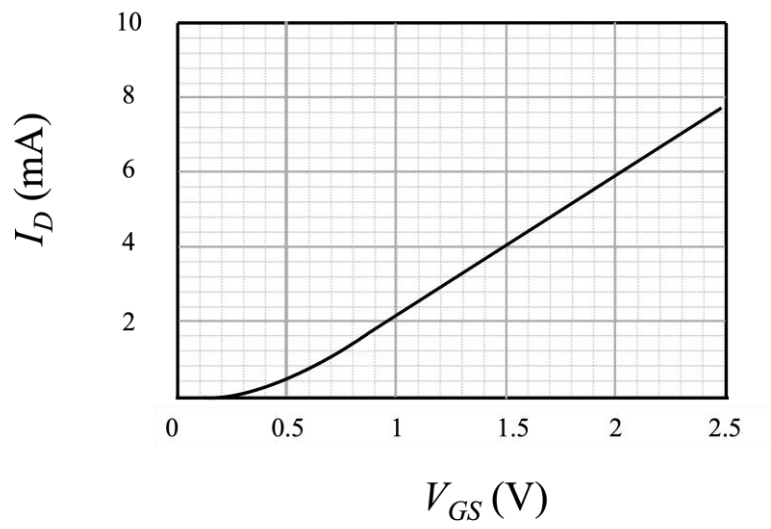


Fig. 3 (a)

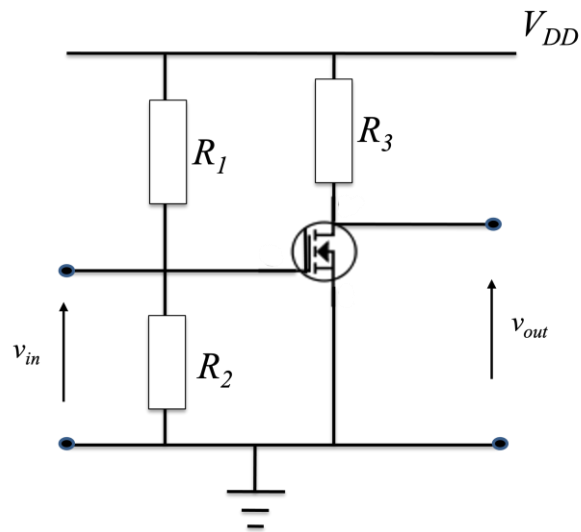


Fig. 3 (b)